

Two Machine Learning Examples

This PDF saves the two examples discussed: Isolation Forest anomaly detection and Apriori association rule mining.

Example 1: Isolation Forest

Purpose: Isolation Forest is an unsupervised algorithm for anomaly or outlier detection. It isolates unusual points faster than normal points by repeatedly making random splits.

Core idea: anomalies are rare and different, so they usually become alone in fewer splits. The number of splits needed to isolate a point is called its **path length**.

Small dataset

Point	Spend Amount	Distance from Home	Type
A	\$20	2 km	Normal
B	\$25	5 km	Normal
C	\$30	3 km	Normal
X	\$500	80 km	Anomaly

Step-by-step tree split

Round	Random feature and split	Result	Path length
1	Feature: Spend Amount. Split: \$150.	A, B, C go left because spend < \$150. X goes right and becomes alone.	X = 1
2	Feature: Distance from Home. Split: 4 km, applied only to A, B, C.	A and C go left. B goes right and becomes alone.	B = 2
3	Feature: Spend Amount. Split: \$26, applied only to A and C.	A goes left. C goes right. Both are now isolated.	A = 3, C = 3

Summary

Data Point	Description	Splits Needed
X	Anomaly: \$500 spend and 80 km away.	1 split - very short
B	Normal: \$25 spend and 5 km away.	2 splits
A	Normal: \$20 spend and 2 km away.	3 splits
C	Normal: \$30 spend and 3 km away.	3 splits

Conclusion: Point X is isolated much faster than the normal points, so Isolation Forest gives it a higher anomaly score. In real use, the algorithm builds many random trees and averages the path lengths.

Example 2: Apriori Algorithm

Purpose: Apriori is an unsupervised association rule mining algorithm. It finds frequent item combinations and rules such as: if a customer buys diapers, they may also buy beer.

Core idea: if an itemset is frequent, all of its subsets must also be frequent. If an item or pair is not frequent, larger combinations containing it can be skipped.

Transaction dataset

Transaction	Items
T1	Bread, Milk
T2	Bread, Diaper, Beer, Eggs
T3	Milk, Diaper, Beer, Cola
T4	Bread, Milk, Diaper, Beer
T5	Bread, Milk, Diaper, Cola

Minimum support: 60%, meaning an itemset must appear in at least 3 out of 5 transactions.

Step 1: Frequent 1-itemsets

Item	Count	Support	Decision
Bread	4	80%	Keep
Milk	4	80%	Keep
Diaper	4	80%	Keep
Beer	3	60%	Keep
Eggs	1	20%	Drop
Cola	2	40%	Drop

Step 2: Frequent 2-itemsets

Item Pair	Count	Support	Decision
{Bread, Milk}	3	60%	Keep
{Bread, Diaper}	3	60%	Keep
{Bread, Beer}	2	40%	Drop
{Milk, Diaper}	3	60%	Keep
{Milk, Beer}	2	40%	Drop
{Diaper, Beer}	3	60%	Keep

Step 3: Check 3-itemsets

The candidate {Bread, Milk, Diaper} is checked because all of its 2-item subsets survived: {Bread, Milk}, {Bread, Diaper}, and {Milk, Diaper}.

It appears in T4 and T5 only, so support = $2 / 5 = 40\%$. Since 40% is below the 60% threshold, it is dropped. No 3-itemsets survive, so Apriori stops growing itemsets.

Step 4: Generate a rule using confidence

From the frequent pair {Diaper, Beer}, we can generate the rule:

Diaper -> Beer

Confidence = transactions with both Diaper and Beer / transactions with Diaper = $3 / 4 = 75\%$.

Final interpretation: in this small dataset, 75% of customers who bought diapers also bought beer. This is the type of pattern supermarkets can use for product placement or recommendations.

Quick comparison

Algorithm	Category	Goal	Output
Isolation Forest	Anomaly detection	Find rare or unusual points	Anomaly score / path length
Apriori	Association rule mining	Find frequent item patterns	Rules like $X \rightarrow Y$, support, confidence, lift